

JOONWON LEE

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SUMMARY

Statistics Ph.D. candidate and data scientist specializing in regression, hypothesis testing, predictive modeling, and statistical measurement for large, complex datasets. Builds reproducible Python workflows across AWS and HPC environments, connecting data preparation, model validation, and interpretable analysis to business decisions. Experience translating pricing and risk questions into measurable outcomes and actionable recommendations for stakeholders.

EDUCATION

Ph.D. in Statistics, Rutgers University, Piscataway, NJ

Expected Dec 2026 | GPA: 3.8/4.0

M.S. in Statistics, University of Minnesota, Minneapolis, MN

Sep 2019 - Sep 2021 | GPA: 3.7/4.0

TECHNICAL SKILLS

Programming and Platforms: Python (NumPy, Pandas, SciPy, PyTorch, LightGBM, scikit-learn), SQL/MySQL, R, Linux, Git, AWS EC2, HPC/SLURM

Statistical Analysis and Machine Learning: Regression (GLM/GLS), predictive modeling, LightGBM, hypothesis testing, experimental design, sequential testing, Monte Carlo simulation, model selection and validation, feature attribution, missing-data analysis

EXPERIENCE

JPMorgan Chase - Summer Quantitative Analytics Associate, Quantitative Finance / Risk Modeling

Summer 2026

- Developed and validated a Sequential Probability Ratio Test (SPRT) framework for binary risk indicators, translating a business monitoring problem into testable hypotheses and measurable decision rules.
- Designed simulation studies to evaluate operating characteristics including Type I/II error, detection delay, and early-decision tradeoffs; communicated model behavior and limitations.

Travelers - Data Science Leadership Program, Business Insurance Pricing Team

Summer 2025

- Developed an end-to-end LightGBM pricing pipeline across 2.46M+ records, including data preparation, feature processing, generalized linear regression benchmarking, model validation, and performance evaluation on AWS EC2.
- Presented model diagnostics and feature-attribution findings as pricing and risk-segmentation recommendations, translating technical results into clear actions for business stakeholders.

RESEARCH AND STATISTICAL ENGINEERING

Ph.D. Dissertation Research - Large-Scale Statistical Modeling and Diagnostics

Sep 2024 - Present

- **Regression and Estimation:** Profiled regression mean structures using generalized least squares and calibrated flexible Gaussian-process models for large, dependent datasets.
- **Data Preparation and Quality:** Built workflows for observation masking, quality filtering, nearest-neighbor construction, dependency-aware ordering, and irregular-grid resampling for noisy, partially observed satellite data.
- **Measurement and Model Evaluation:** Designed scale- and frequency-resolved tests to identify misspecification and quantify how missingness, noise, and dependence affect statistical estimates.
- **Scalable Computing:** Implemented reusable Python/PyTorch pipelines with GLS profiling, L-BFGS optimization, CPU/GPU execution, cached artifacts, and restartable HPC jobs for up to 145,008 observations per day.

ADDITIONAL

Coursework: Econometrics, Machine Learning, Probability Theory, Linear Algebra, Statistical Computing, Stochastic Processes, Advanced Theory of Statistics I-II, Data Structures and Algorithms, Microeconomics

Languages: Korean (Native); English (Fluent); Chinese (Basic)